CSE 423 Quiz - 4

Set: A

Mark: 20+2, Time: 20 minutes

Name: ID: Section:

1. Convert CMY (0.2, 0.65, 0.78) to HSV 8

$R = 1 - 0.2 = 0.8$

$G = 1 - 0.65 = 0.35$

$B = 1 - 0.78 = 0.22$

$c_{max} = 0.8$ (R)

$c_{min} = 0.22$ (B)

$diff = 0.58$

$$h = \frac{g - b}{diff} \times 60 = 13.44^\circ$$

$$S = \frac{diff}{c_{max}} = 0.73$$

$$V = c_{max} = 0.8$$

HSV (13.44°, 0.73, 0.8)

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2. A piecewise spline, f(t), is defined as:

$$f(t) = (1.5t^2 - 0.5, 4t^{-3}) \text{ when } t \leq 1$$
$$f(t) = (t^3, 1 + 3t^{-4}) \text{ when } t > 1$$

Find the maximum value of n for G(n) and C(n) continuity separately at t = 1. 0. Analyze whether the splines join smoothly by examining their continuity conditions. 14

$\lim_{t \to 1^-} f(t) = (1.5t^2 - 0.5, 4t^{-3})$

$= (1.5 \times 1^2 - 0.5, 4 \times 1^{-3})$

$= (1, 4)$

$\lim_{t \to 1^+} f(t) = (t^3, 1 + 3t^{-4})$

$= (1, 4)$

$\lim_{t \to 1^-} f(t) = \lim_{t \to 1^+} f(t) = f(1)$

$\therefore$ C(0) and G(0)

$f'(t) = (3t, -12t^{-4}) \quad t \leq 1$

$f'(t) = (3t^2, -12t^{-5}) \quad t > 1$

$\lim_{t \to 1^-} f'(t) = (3t, -12t^{-4}) = (3, -12) = f'(1)$

$\lim_{t \to 1^+} f'(t) = (3t^2, -12t^{-5}) = (3, -12)$

$\therefore \lim_{t \to 1^-} f'(t) = \lim_{t \to 1^+} f'(t) = f(1)$

$\therefore$ C(1) and G(1)

$f''(t) = (3, +48t^{-5}) \quad t \leq 1$

$f''(t) = (6t, 60t^{-6}) \quad t > 1$

$\lim_{t \to 1^-} f''(t) = (3, 48t^{-5})$

$= (3, 48)$

$\lim_{t \to 1^+} f''(t) = (6t, 60t^{-6})$

$= (6, 60)$

$\lim_{t \to 1^-} f''(t) \neq \lim_{t \to 1^+} f''(t)$

$\therefore$ not C(2). Also not G(2)

CSE 423 Quiz - 4

Set: B

Mark: 20+2, Time: 20 minutes

Name: ID: Section:

1. Convert HSV (25°, 0.6, 0.85) to CMY 8

$$c = V \times S = 0.85 \times 0.6 = 0.51$$

$$X = c \times \left(1 - \left|(H/60 \bmod 2) - 1\right|\right)$$

$$= 0.51 \times \left(1 - \left|\left(\tfrac{25}{60} \bmod 2\right) - 1\right|\right)$$

$$= 0.51 \times \left(1 - \left|(0.42 \bmod 2) - 1\right|\right)$$

$$= 0.51 \times (1 - |0.42 - 1|)$$

$$= 0.51 \times 0.42$$

$$= 0.21$$

$$m = v - c = 0.85 - 0.51 = 0.34$$

$0° \le H \le 60°$ $\therefore$ $R = c = 0.51$ $G = X = 0.21$ $B = 0$

$R' = R + m = 0.85$ $G' = G + m = 0.55$ $B' = B + m = 0.34$

$C = 1 - 0.85 = 0.15$ $M = 1 - 0.55 = 0.45$ $Y = 1 - 0.34 = 0.66$

CMY (0.15, 0.45, 0.66)

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2. A piecewise spline, f(t), is defined as:

$$f(t) = (1.5t^2 - 0.5, 4t^{-3}) \text{ when } t \leq 1$$
$$f(t) = (t^3, 1 + 3t^{-4}) \text{ when } t > 1$$

Find the maximum value of n for G(n) and C(n) continuity separately at t = 1. 0. Analyze whether the splines join smoothly by examining their continuity conditions. 14

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